



Internship Project

#3 Advanced Equity Portfolio Risk Management: VaR, Expected Shortfall, GARCH Modeling & Backtesting

Objective:

Build a Python-based risk analytics platform to calculate, scale, backtest, and visualize **Value at Risk** for an equity portfolio using:

- Historical Simulation VaR
- EWMA Volatility Scaling
- GARCH Volatility Scaling
- GJR-GARCH Volatility Scaling
- VaR Backtesting

Project Scope

Portfolio Type

- Equity portfolio
- Long-only portfolio
- Long-short portfolio optional
- Multi-stock portfolio
- Index benchmark comparison optional

Risk Measures

- 1-day VaR
- 10-day VaR
- Expected Shortfall
- Rolling VaR
- Portfolio volatility
- Drawdown
- Backtesting exceptions



Phase 1 – Data Collection

Collect

- Daily adjusted close prices
- Daily returns
- Portfolio weights
- Market index prices
- Sector information optional
- Risk-free rate optional

Data Sources

- Yahoo Finance
- Stooq
- Alpha Vantage
- Polygon
- WRDS, if available

Example Universe

- AAPL
- MSFT
- NVDA
- JPM
- XOM
- AMZN
- META
- GOOGL
- TSLA
- UNH



Phase 2 – Data Cleaning

Tasks

- Adjust prices for splits and dividends
- Align trading dates across stocks
- Handle missing values
- Remove stale prices
- Winsorize extreme return errors if needed
- Convert prices into log returns
- Build weighted portfolio returns

Output

- Clean price matrix
- Clean return matrix
- Portfolio return series
- Rolling volatility series

Phase 3 – Portfolio Construction

Inputs

- Tickers
- Weights
- Initial portfolio value
- Rebalancing frequency

Portfolio Options

- Equal-weighted portfolio
- User-defined weights
- Market-cap weighted portfolio
- Long-short portfolio optional

Output

- Daily portfolio value
- Daily portfolio P&L
- Daily portfolio returns



Phase 4 – Historical Simulation VaR

Implement

- Rolling historical window
- 250-day window
- 500-day window
- 750-day window
- Confidence levels:
 - 95%
 - 99%
 - 99.5%

Calculate

- Portfolio P&L distribution
- Historical Simulation VaR
- Historical Expected Shortfall
- Rolling VaR time series

Charts

- Historical return distribution
- VaR threshold line
- Rolling VaR chart
- Tail loss chart
- VaR vs Expected Shortfall chart

Phase 5 – EWMA Volatility Scaling

Model

- Exponentially Weighted Moving Average
- Common lambda:
 - 0.94 for daily returns

Implement

- EWMA variance
- EWMA volatility
- Scale historical returns by current volatility
- Calculate volatility-scaled Historical VaR

Formula Logic



- Estimate volatility for each day
- Rescale old returns to today's volatility
- Give more realistic VaR during high-volatility regimes

Charts

- EWMA volatility chart
- Raw returns vs volatility-scaled returns
- Historical VaR vs EWMA-scaled VaR
- Exception comparison chart

Phase 6 – GARCH Volatility Scaling

Model

- GARCH(1,1)

Implement

- Fit GARCH model on portfolio returns
- Estimate conditional volatility
- Forecast 1-day ahead volatility
- Scale historical returns using GARCH volatility
- Calculate GARCH-scaled Historical VaR

Output

- Conditional volatility
- Forecast volatility
- Scaled returns
- GARCH-scaled VaR

Charts

- GARCH conditional volatility
- GARCH forecast volatility
- Historical VaR vs GARCH-scaled VaR
- GARCH standardized residuals
- Volatility clustering chart



Phase 7 – GJR-GARCH Volatility Scaling

Model

- GJR-GARCH

Why Use It

- Captures asymmetric volatility
- Negative returns can increase future volatility more than positive returns
- Useful for equity portfolios

Implement

- Fit GJR-GARCH model
- Estimate conditional volatility
- Forecast next-day volatility
- Scale historical returns using GJR-GARCH volatility
- Calculate GJR-GARCH-scaled Historical VaR

Output

- Conditional volatility
- Leverage effect parameter
- Forecast volatility
- GJR-GARCH-scaled VaR

Charts

- GJR-GARCH volatility chart
- GARCH vs GJR-GARCH volatility comparison
- Leverage effect visualization
- Historical VaR vs GJR-GARCH-scaled VaR



Phase 8 – VaR Backtesting

Objective

Check whether VaR model performs correctly.

Backtesting Window

- Rolling out-of-sample period
- Example:
 - First 750 days for estimation
 - Remaining days for backtesting

Exceptions

- Exception occurs when:
 - Actual loss > VaR

Backtesting Tests

- Kupiec POF Test
- Christoffersen Independence Test
- Christoffersen Conditional Coverage Test
- Basel Traffic Light Test

Outputs

- Number of exceptions
- Expected exceptions
- Exception ratio
- P-value
- Green / Yellow / Red zone
- Pass / Fail conclusion



Phase 9 – Model Comparison

Compare

- Historical Simulation VaR
- EWMA-scaled Historical VaR
- GARCH-scaled Historical VaR
- GJR-GARCH-scaled Historical VaR

Metrics

- Number of exceptions
- Exception ratio
- Average VaR
- VaR volatility
- Worst daily loss
- Expected Shortfall
- Backtesting p-values
- Traffic light zone

Charts

- VaR model comparison line chart
- Actual P&L vs VaR chart
- Exceptions overlay chart
- Backtesting summary bar chart
- Model ranking table



Phase 10 – Stress Testing

Historical Stress Scenarios

- 2008 Financial Crisis
- COVID Crash 2020
- Inflation shock 2022
- Banking stress 2023
- Tech selloff scenario

Hypothetical Scenarios

- Equity market down 5%
- Equity market down 10%
- Equity market down 20%
- Volatility doubles
- Correlation spikes
- Sector-specific shock

Outputs

- Stressed portfolio loss
- Stressed VaR
- Stressed Expected Shortfall
- Worst-case loss
- Scenario ranking

Charts to Build

Return & P&L Charts

- Portfolio value over time
- Daily portfolio returns
- Daily portfolio P&L
- Drawdown chart
- Return distribution

VaR Charts

- Rolling Historical VaR
- EWMA-scaled VaR
- GARCH-scaled VaR
- GJR-GARCH-scaled VaR
- VaR vs Actual Loss



- VaR vs Expected Shortfall

Volatility Charts

- Rolling volatility
- EWMA volatility
- GARCH volatility
- GJR-GARCH volatility
- Volatility regime comparison

Backtesting Charts

- Exception timeline
- Exceptions by model
- Basel Traffic Light chart
- Actual P&L vs VaR
- Expected vs Actual exceptions

Stress Testing Charts

- Scenario loss bar chart
- Stress VaR comparison
- Sector shock impact
- Worst loss ranking

Dashboard Requirements

Build Using

- Streamlit
- Plotly
- Dash optional

Dashboard Pages

Portfolio Page

- Ticker input
- Weight input
- Portfolio value input
- Date range selector
- Portfolio return chart



VaR Page

- Confidence level selector
- Historical window selector
- VaR model selector
- VaR output
- Expected Shortfall output

Volatility Scaling Page

- EWMA lambda input
- GARCH model output
- GJR-GARCH model output
- Volatility comparison chart

Backtesting Page

- Actual P&L vs VaR
- Exceptions table
- Kupiec test result
- Christoffersen test result
- Basel Traffic Light result

Stress Testing Page

- Scenario selector
- Stress loss output
- Stress VaR output
- Scenario comparison chart

Validation Checks

Data Validation

- No missing return dates
- Weights sum to 100%
- No look-ahead bias
- Returns correctly aligned
- Portfolio P&L calculated correctly

VaR Validation

- VaR is positive loss number
- More confidence level gives larger VaR



- 10-day VaR > 1-day VaR
- Expected Shortfall > VaR
- VaR increases during high volatility

Backtesting Validation

- Expected exception rate close to selected confidence level
- Exceptions not clustered heavily
- Kupiec test implemented correctly
- Christoffersen test implemented correctly

Final Deliverables

Code

- Clean Python package
- Data pipeline
- VaR engine
- Volatility scaling engine
- Backtesting engine
- Stress testing module
- Unit tests

Dashboard

- Interactive risk dashboard
- Portfolio inputs
- VaR outputs
- Backtesting outputs
- Stress testing outputs

Report

- Executive summary
- Data description
- Portfolio construction
- Historical Simulation VaR methodology
- EWMA methodology
- GARCH methodology
- GJR-GARCH methodology
- Backtesting results
- Model comparison
- Stress testing
- Limitations



- Future improvements

Presentation

- 10–15 slides
- Dashboard demo
- VaR comparison
- Backtesting conclusion
- Risk management insights

Expected Learning Outcomes

- Equity portfolio risk modeling
- Historical Simulation VaR
- Volatility scaling
- EWMA volatility
- GARCH modeling
- GJR-GARCH leverage effect
- Expected Shortfall
- Backtesting exceptions
- Kupiec test
- Christoffersen test
- Basel Traffic Light framework
- Stress testing
- Risk dashboard development

Organizations where similar workflows are commonly used include consulting firms such as **Deloitte, PwC, KPMG, and EY**; banks such as **JPMorgan Chase, Goldman Sachs, Morgan Stanley, Bank of America, and Citigroup**; asset managers such as **BlackRock and PIMCO**;